

Deep Temporal-Spatial Feature Learning for Motor Imagery-Based Brain–Computer Interfaces

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Abstract—Motor imagery (MI) decoding is an important part of brain-computer interface (BCI) research, which translates the subject's intentions into commands that external devices can execute. The traditional methods for discriminative feature extraction, such as common spatial pattern (CSP) and filter bank common spatial pattern (FBCSP), have only focused on the energy features of the electroencephalography (EEG) and thus ignored the further exploration of temporal information. However, the temporal information of spatially filtered EEG may be critical to the performance improvement of MI decoding. In this paper, we proposed a deep learning approach termed filter-bank spatial filtering and temporal-spatial convolutional neural network (FBSF-TSCNN) for MI decoding, where the FBSF block transforms the raw EEG signals into an appropriate intermediate EEG presentation, and then the TSCNN block decodes the intermediate EEG signals. Moreover, a novel stage-wise training strategy is proposed to mitigate the difficult optimization problem of the TSCNN block in the case of insufficient training samples. Firstly, the feature extraction layers are trained by optimization of the triplet loss. Then, the classification layers are trained by optimization of the cross-entropy loss. Finally, the entire network (TSCNN) is fine-tuned by the back-propagation (BP) algorithm. Experimental evaluations on the BCI IV 2a and SMR-BCI datasets reveal that the proposed stage-wise training strategy yields significant performance improvement compared with the conventional end-to-end training strategy, and the proposed approach is comparable with the state-of-the-art method.

Index Terms—Motor imagery (MI), electroencephalography (EEG), deep learning, convolutional neural network (CNN), triplet loss.

I. INTRODUCTION

BRAIN-computer interface (BCI) system provides a new communication pathway between brain and external devices without relying on the regular brain nerves and muscles. It can help the disabled to communicate with others, or provide man-machine interaction for the healthy

people [1], [2]. Among the non-invasive BCI systems, motor imagery (MI) is attracting increasing attention due to its advantage of no external stimulus required [3]–[5]. In such an MI-based BCI system, the subject is asked to imagine moving different parts of their body to trigger the neuronal activities in specific brain areas that are related to the movements. During the MI process, the event-related desynchronization (ERD) and event-related synchronization (ERS) patterns of electroencephalography (EEG) occur simultaneously and reflect event-related changes of the brain activities in specific frequency bands to a great extent [6]–[9]. The pattern of ERD/ERS generated by the imagined movements is similar to that of actual movements [10], [11], but varies among different subjects. Thus intensive training is required for each subject [2]. Besides, the EEG recordings usually have a low signal-to-noise ratio (SNR), which makes it challenging to decode the recorded EEG of these neuronal activities correctly. Therefore, the efficient feature extraction and correct classification are two major challenges in the MI-based BCI system.

Existing representative methods, such as common spatial filter (CSP) [12] and its extensions [4], [13], are used frequently as spatial filters to capture the ERD/ERS patterns of MI EEG. Especially, the CSP not only extracts the discriminative energy features of EEG signals but also provides the weight knowledge of the EEG channels. Taking the two-class MI tasks as an example, the CSP aims to seek a set of linear transformations (i.e., spatial filters) that maximize the variance of one class while minimizing the variance of the other. After spatial filtering of EEG, the logarithmic energies of the spatially filtered channels are calculated and taken as features for the follow-up classification. However, the CSP is quite sensitive to the band-pass frequency range, and thus the classification performance may vary dramatically when the choice of the frequency band changes [14]. Although a wide frequency band of 8–30 Hz was suggested, studies proved that selecting subject-specific frequency bands could improve the classification performance of the MI-based BCI [15]. The traditional method is to perform an exhaustive search and manual adjustment for each subject, which is time-consuming. To address this issue, the filter bank common spatial pattern (FBCSP) [16], [17] method was proposed which divides a wide frequency band into multiple smaller frequency bands and then performs an autonomous selection of discriminative CSP features from these frequency

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bands via mutual information-based best individual feature (MIBIF) [16]. After feature extraction of EEG signals, many classical classifiers, such as naive bayesian parzen window (NBPW) [16], linear discriminant analysis (LDA) [18], and support vector machine (SVM) [19], can be applied to classify these energy features. The research in [16] showed that FBCSP features are more separable than CSP features when using the same classifier. However, both CSP and FBCSP ignore the further exploration of temporal information of spatially filtered EEG.

Recently, the deep learning methods have achieved state-of-the-art performance in the fields of computer vision and speech recognition [20]. However, the application of deep learning in MI decoding is relatively scarce, for the non-stationary nature of brain activity and the low SNR of recorded EEG signals make it challenging to design an ideal deep network framework [21], [22]. By borrowing the idea from deep learning in speech recognition [23], it is feasible to transform the raw EEG signals into an appropriate intermediate representation without any significant loss in discriminative information for the follow-up deep learning. In general, the common EEG transformation includes spatial filtering (SF), fast Fourier transformation (FFT), wavelet-packet decomposition [24], *etc.* After the EEG transformation, a deep neural network architecture is designed to decode the intermediate EEG representation.

Some representative deep learning methods for MI decoding were proposed in literature [25]–[28]. An *et al.* [25] proposed a boosted deep belief networks (DBNs) structure, in which the frequency components of EEG signals obtained by FFT were fed to some single-channel DBNs for training and finally these weak classifiers were boosted to a powerful one via Ada-boost. The results showed that the DBN classifier demonstrated better performance than SVM when using the same frequency components. Yang *et al.* [26] proposed a frequency complementary map selection (FCMS) scheme for the convolutional neural network (CNN) based on the augmented CSP (ACSP) to reduce the dependency of the feature maps among frequency bands. However, it failed to exploit the temporal information of EEG. Sakhavi *et al.* [27] designed CNN to learn the temporal information of EEG, where the input to the network is the envelope representation of the filtered EEG via Hilbert transformation. Since the envelope signal is of low-frequency nature, it could be down-sampled, and hence the dimensionality of the data is lowered. Schirrmeister *et al.* [28] proposed a deep convolutional network (Deep ConvNet) architecture for EEG decoding, which slid the input window within a single trial to get multiple crops for the network training and prediction. However, the increasing depth of the ConvNet results in the more parameters to be tuned, and the network may tend to be over-fitting due to the insufficient training samples. Data augmentation is an effective method to solve the problem of data shortage, where the generated data can be obtained by the geometric transformations, including shift, scale, and rotation to the real data [29]. Unfortunately, these geometric transformations of EEG may work badly due to the non-stationary nature of the EEG signals [30]. For recent advances and new frontiers in deep learning-based BCI systems, readers are referred to [31] for more details.

In this paper, we proposed a deep model termed filter-bank spatial filtering and temporal-spatial convolutional neural network (FBSF-TSCNN) for motor imagery decoding. Specifically, the FBSF block, which transforms the raw EEG signals into an appropriate intermediate EEG presentation, is derived from the FBCSP algorithm. Both FBSF and FBCSP have the same processing steps, including band-pass filtering through filter bank, spatial filtering, and MIBIF selection. The differences are that the FBSF returns the intermediate (spatially filtered) EEG signals with temporal representation, while the FBCSP only returns logarithmic energies of the spatially filtered channels. Since the intermediate EEG signals provided by the FBSF still keep the temporal representation, the subsequent deep model can further exploit the temporal information. Then, the TSCNN block learns the discriminative features of the intermediate EEG signals and returns the decoding results. However, for many reasons, the MI EEG dataset available for training is frequently small in sample size, which may result in the deep model susceptible to poor local minima during gradient-based optimization starting from random initialization. Motivated by the success of the triplet mining method in face recognition [32], we proposed a novel stage-wise training strategy for the TSCNN block in contrast to the conventional end-to-end training strategy which simply optimizes the cross-entropy loss. Firstly, the feature extraction layers are trained by optimization of the triplet loss. Then, the classification layers are trained by optimization of the cross-entropy loss. Finally, the entire network is fine-tuned by the back-propagation (BP) algorithm. Compared with the conventional end-to-end training strategy, the proposed stage-wise training strategy could yield superior classification performance for the proposed model in MI tasks.

The major contributions of this paper are threefold.

- 1) The intermediate EEG signals provided by the FBSF still keep the temporal representation, so the subsequent deep model can further exploit the temporal information.
- 2) The TSCNN block is proposed to decode the intermediate EEG signals, and it can provide insightful observation into the discriminative temporal information.
- 3) A novel stage-wise training strategy is proposed to train the TSCNN block, and it yields superior performance for the model compared with the end-to-end training strategy, as supported by numerical results.

The remainder of the paper is organized as follows. Section II presents the framework of the proposed approach. Section III describes the experimental settings, and then shows the results and discussion. Some analysis is given in Section IV, and the conclusions are drawn in Section V.

II. METHODOLOGY

In this section, we first introduced the basic procedures of spatial filtering. Then we explained the basic blocks of the proposed approach, including the FBSF block and TSCNN block. Next, we described the proposed stage-wise training strategy for the TSCNN block. Finally, we summarized the proposed approach for motor imagery decoding.

A. Introduction of Spatial Filtering

The spatial filtering (SF) method derived from the standard CSP [12] algorithm is commonly used to transform the raw MI EEG signals. The procedures of SF are as follows:

- 1) Filter the raw EEG $S \in \mathbb{R}^{ch \times T}$ using a band-pass filter to obtain filtered EEG $E \in \mathbb{R}^{ch \times T}$, where ch and T are the number of the recorded EEG channels and the number of the single-channel sampling points respectively;
- 2) Calculate the spatial filter matrix $W_b \in \mathbb{R}^{ch \times ch}$ by maximizing the following objective function

$$J(w) = \frac{w^T \Sigma_1 w}{w^T \Sigma_2 w} \quad (1)$$

under the constraint

$$w^T \Sigma_1 w + w^T \Sigma_2 w = 1 \quad (2)$$

where Σ_1 and Σ_2 are estimates of the covariance matrices of the band-pass filtered EEG regarding two classes of MI tasks respectively, and $w \in \mathbb{R}^{ch \times 1}$ is a column vector that forms the spatial filter matrix W_b . This constraint optimization problem can be translated into solving the eigenvalue decomposition problem [12]

$$\Sigma_1 W_b = (\Sigma_1 + \Sigma_2) W_b D_b \quad (3)$$

where D_b is the diagonal matrix that contains the eigenvalues of Σ_1 ;

- 3) Select the first N_w columns and last N_w columns (corresponding to N_w largest and N_w smallest eigenvalues) of W_b to form a new spatial filter $\overline{W}_b \in \mathbb{R}^{ch \times (2 \times N_w)}$;
- 4) Transform the band-pass filtered EEG E by multiplying $\overline{W}_b$ to obtain the spatially filtered output G , i.e.,

$$G = \overline{W}_b^T E \quad (4)$$

where $G \in \mathbb{R}^{(2 \times N_w) \times T}$;

- 5) In the case of multi-class tasks, a one-versus-rest (OVR) strategy is appointed here. Suppose there are k ($k \geq 3$) classes of MI tasks, it will lead to a total of $c = k \times 2 \times N_w$ spatially filtered channels.

From the procedures above, it can be found that the SF performs the EEG channel projection, which extracts the dependencies of the channels without transforming the temporal representation of EEG.

B. Derived Filter Bank Spatial Filtering

For the SF method, the band-pass frequency range needs to be set manually before spatial filtering. In this section, the filter bank spatial filtering (FBSF), which is derived from the FBCSP [16], is proposed to transform the raw EEG signals into an appropriate intermediate EEG representation. The procedures of the FBSF are as follows:

- 1) Filter the raw EEG $S \in \mathbb{R}^{ch \times T}$ through a filter bank comprising r different band-pass filters, and the corresponding output for the b^{th} frequency band is denoted as $E_b \in \mathbb{R}^{ch \times T}$;
- 2) Calculate the spatial filter matrix $W_b \in \mathbb{R}^{ch \times ch}$ for the band-pass filtered EEG E_b by (3);

- 3) Select the first N_w columns and last N_w columns (corresponding to N_w largest and N_w smallest eigenvalues) of W_b to form a new spatial filter $\overline{W}_b \in \mathbb{R}^{ch \times (2 \times N_w)}$;
- 4) Perform spatial filtering of E_b to obtain the corresponding output $G_b \in \mathbb{R}^{(2 \times N_w) \times T}$ by (4);
- 5) Calculate the energy ($R_{b,i}$) of i^{th} channel of G_b , then normalize it ($\overline{R}_{b,i} = R_{b,i} / (\sum_{j=1}^{2 \times N_w} R_{b,j})$), and take the logarithmic format ($L_{b,i} = \log \overline{R}_{b,i}$);
- 6) Concatenate these logarithmic energies from all frequency bands and perform the MIBIF [16] algorithm to select N_s spatially filtered channels and their pairs;
- 7) In the case of multi-class tasks, the OVR strategy is appointed here. Suppose there are k ($k \geq 3$) classes of MI tasks, it will lead to a total of $c = k \times 2 \times N_s$ spatially filtered channels.

The intermediate EEG provided by the FBSF still keep the temporal representation, which can be inputted to the follow-up TSCNN block for further decoding. Since EEG is composed of different types of oscillatory activities, where oscillations in the alpha and beta bands are particularly important for distinguishing between different brain states during motor imagery [33]. In this paper, the filter bank in FBSF is set up with reference to [34], and will be described in Section III-A.

C. Proposed Temporal-Spatial Convolutional Neural Network

As shown in Fig. 1(b), the proposed TSCNN block, consisting of the feature extraction layers and the classification layers, is used to decode the intermediate EEG.

1) Design of the Feature Extraction Layers of the TSCNN: This sub-block consists of one temporal convolutional layer of 16 kernels, one temporal convolutional layer of 32 kernels, one spatial convolutional layer of 32 kernels, one average pooling layer of 32 kernels, and one flattening layer. Furthermore, the batch normalization ((BN)) [35] modules are inserted after the convolutional layers, and the rectified linear units (ReLU) [36] followed by the dropout operation are inserted after the former two BN modules.

The tensor flows involved in this sub-block are as follows. At first, the intermediate EEG trial with the shape of $(1, c, 281)$ is inputted into the first convolutional layers with 16 kernels of size $(1, 20)$ to learn the temporal features along the time dimension. Then, the second convolutional layer with 32 kernels of size $(1, 15)$ further learns the temporal features along the time dimension. Next, the spatial convolutional layer with 32 kernels of size $(c, 1)$ fuses the temporal features from all channels. After these convolutional operations, the average pooling layer with 32 kernels is applied to reduce the length of outputs in the time dimension and obtain more robust features. Finally, a total of 32 feature maps is flattened into one 1-D vector $f \in \mathbb{R}^{256}$. Besides, the dropout operation acts with a probability of 50% to combat over-fitting problem during training while simply computing an identity function during the evaluation, and the BN modules after the convolutional layers make the feature maps normally distributed, which

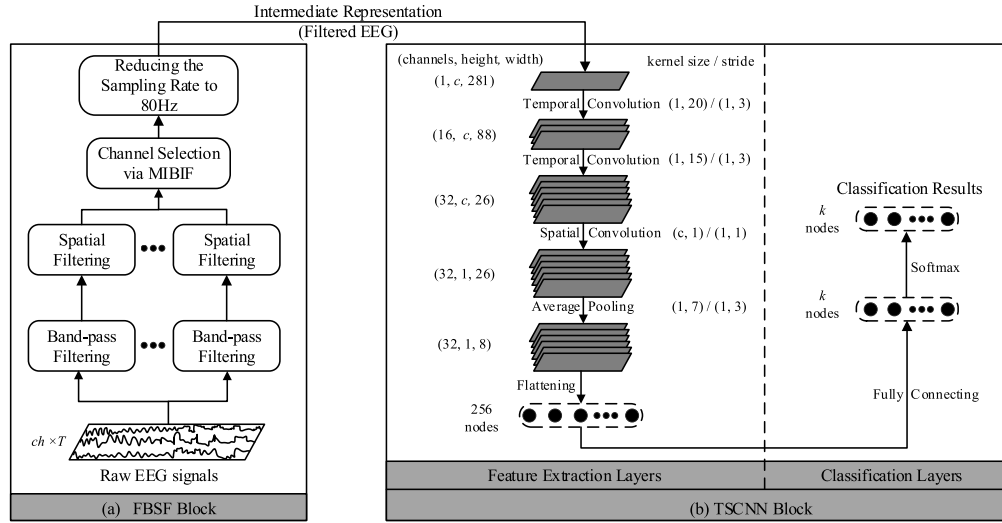


Fig. 1. An schematic illustration of the FBSF-TSCNN architecture. (a) FBSF block for data transformation; (b) TSCNN block for decoding the intermediate EEG. For BCI IV 2a dataset, parameters $ch = 22$, $T = 876$, $c = 32$, $k = 4$; for SMR-BCI dataset, parameters $ch = 15$, $T = 1793$, $c = 8$, $k = 2$.

contributes to tuning the optimal parameters of the deep networks.

2) *Design of the Classification Layers of the TSCNN*: This sub-block consists of one fully connected layer and one softmax layer, which aims to achieve the final decoding results. The 1-D feature vector f provided by the feature extraction layers, is fed to the full connected layer, and then the softmax layer transforms the predicted output into conditional probabilities. The label with the highest probability is decided as the final classification result of the TSCNN.

D. Proposed Stage-Wise Training Strategy

For many reasons, the MI EEG dataset available for training is frequently small in sample size, and hence the gradient-based optimization for the deep model with random initialization appears to get stuck in poor solutions. To mitigate the difficult optimization problem of deep networks, we can try initialing the weights of hidden layers in a region near good local minima. By borrowing the triplet mining method in [32], we developed a novel stage-wise training strategy for the proposed TSCNN block. In contrast to the end-to-end training strategy which simply optimizes the cross-entropy loss, the proposed stage-wise training strategy includes three stages.

In the first stage, the feature extraction layers are trained by optimization of the triplet loss [32]. Suppose that the feature vector f_i of anchor input trial X_i should be closer to f_i^+ of positive input trial X_i^+ with same label than to f_i^- of negative input trial X_i^- with different label, i.e.,

$$\|f_i - f_i^+\|_2^2 + \alpha \leq \|f_i - f_i^-\|_2^2 \quad (5)$$

where α is a margin that is enforced between positive and negative pairs. Therefore, the triplet loss function that should be minimized is defined as follow:

$$L_1 = \sum_{i=1}^N \max(\|f_i - f_i^+\|_2^2 + \alpha - \|f_i - f_i^-\|_2^2, 0) \quad (6)$$

Note that if all the possible input triplets (X_i, X_i^+, X_i^-) are used for training, it may cause the network to converge slowly or fail to converge. In this paper, we perform the online selection of mini-batch triplets in the training process [32].

In the second stage, only the classification layers are trained by optimization of the cross-entropy loss since the previous feature extraction layers have been trained and the weights are fixed. For multi-class MI tasks, the cross-entropy loss that should be minimized is given as follows,

$$L_2 = - \sum_{i=1}^N \log p[y_i | X_i] \quad (7)$$

where $y_i \in \{1, 2, \dots, k\}$ is the ground true label of the input X_i , $p[y_i | X_i]$ is the normalized probability value that predicts the trial X_i belongs to the class y_i , k is the number of categories of MI tasks, and N is the number of training trials.

In the third stage, we fine-tune the entire network (TSCNN) by the BP algorithm.

In summary, the procedures of the proposed FBSF-TSCNN are shown in Fig. 1. The FBSF block transforms the raw EEG signals into an intermediate EEG presentation by band-pass filtering through a filter bank, spatial filtering, and channel selection via MIBIF. Then, the intermediate EEG signals are inputted to the TSCNN block for further decoding, where the stage-wise training strategy is proposed to train the TSCNN block. Firstly, the features extraction layers are trained by optimization of the triplet loss. Then, the classification layers are trained by optimization of the cross-entropy loss. Finally, the whole TSCNN is fine-tuned by the BP algorithm.

III. NUMERICAL RESULTS

A. Experimental Setup

1) *Data Description*: To verify the performance of the proposed approach for MI decoding, extensive and systematic experiments have been conducted on the BCI IV 2a [37]

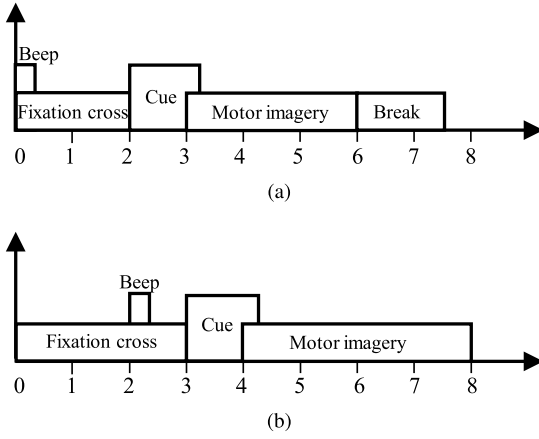


Fig. 2. Time scheme of the paradigm. (a) BCI IV 2a dataset; (b) SMR-BCI dataset.

and SMR-BCI [34] datasets. Both public datasets were provided by Institute for Knowledge Discovery (Laboratory of Brain-Computer Interfaces), Graz University of Technology, and both are freely accessible via <http://www.bnci-horizon-2020.eu/database/data-sets>. The description of these two public datasets are as follows:

- (a) BCI IV 2a dataset includes four-class MI tasks (left hand, right hand, feet, and tongue) obtained from 9 subjects. The EEG was measured with 22 Ag/AgCl electrodes at a sampling rate of 250Hz, and the sensitivity of the amplifier was set to 100 μ V. The data for each subject consists of two sessions: one for training and the other for testing. Each session has 288 trials, with an average of 72 trials for each class. The cue-based BCI paradigm of the MI EEG signals recording is shown in Fig. 2(a). In this paper, the time interval of single-trial EEG data was restricted to the time segment between 2.5s and 6s, leading to 876 sampled data points per channel.
- (b) SMR-BCI dataset includes two-class MI tasks (right hand and feet) obtained from 14 subjects. The EEG was measured with a biosignal amplifier and 15 Ag/AgCl electrodes at a sampling rate of 512Hz. The data for each subject consisted of 100 trials without feedback for training and 60 trials with feedback for testing. The cue-based BCI paradigm of the MI EEG signals recording is shown in Fig. 2(b). In this paper, the time interval of single-trial EEG data was restricted to the time segment between 4.5s and 8s, leading to 1793 sampled data points per channel.

For each subject from two public datasets, we randomly selected 70% (i.e., 200 or 70 trials) of the training data for training and the remaining 30% (i.e., 88 or 30 trials) for validation, and ensured balanced distribution among classes.

2) Methods Evaluated: We evaluated the proposed method FBSF-TSCNN against the following competing methods.

- (a) TSCNN: The raw EEG signals were band-pass filtered between 8 and 30Hz, and then inputted to the TSCNN for MI decoding.
- (b) SF-TSCNN: The SF followed by the TSCNN was applied for MI decoding.

- (c) CSP-NBPW: The CSP followed by the NBPW classifier was applied for MI decoding [16].
- (d) FBCSP-NBPW: The FBCSP followed by the NBPW classifier was applied for MI decoding [16].
- (e) C2CM: A deep learning model proposed in [27], where the input to the network is the envelope representation of the filtered EEG via Hilbert transformation.
- (f) Deep ConvNet: A deep learning model proposed in [28], which slid the input window within a single trial to get multiple crops for the model training and prediction.

Particularly, for the methods of TSCNN, SF-TSCNN, and FBSF-TSCNN, the superscript $^{\circ}$ indicates that the TSCNN block of the model adopts the end-to-end training strategy; otherwise, the TSCNN block of the model adopts the stage-wise training strategy.

3) Parameters Setting: For SF and CSP algorithms, the band-pass frequency range is set to 8-30Hz, and the parameter N_w is set to 4. For FBSF and FBCSP algorithms, the filter bank comprises 12 band-pass filters, i.e., the frequency range of $[i, i + 2]$ Hz and $[j, j + 5]$ Hz, where $i = 6, 8, 10, 12$ and $j = 14, 17, 20, 23, 26, 29, 32, 35$ [34]. Additionally, the parameters N_w and N_s in FBSF and FBCSP algorithms are set to 2 and 4 respectively.

Note that the spatially filtered EEG signals in the FBSF (SF) block have a cutoff frequency below 40 Hz, which means a sampling rate of 80 Hz is sufficient for the signal according to Nyquist sampling theorem, so we reduced the sampling rate to 80Hz to lowered the dimensionality of intermediate EEG representation. Therefore, the dimensions of intermediate EEG (input to the TSCNN block) were (1, 32, 281) and (1, 8, 281) on BCI IV 2a and SMR-BCI datasets, respectively.

In the training process of the TSCNN block, the Adam optimizer [38] is employed with the initial learning rate of 0.001 and betas of (0.9, 0.999).

4) Performance Metrics: In this paper, the Cohen's kappa coefficient (κ) [39] and accuracy are chosen as the performance metrics for these two experimental datasets. The Cohen's kappa coefficient is defined as follows:

$$\kappa = \frac{p_o - p_e}{1 - p_e} \quad (8)$$

where p_o is the ratio of observed agreement that is equal to the average classification accuracy rate over all the classes, p_e is the ratio of expected agreement when both annotators assign labels randomly. And p_e can be computed by the following formula:

$$p_e = \frac{\mathbf{m}_c \times \mathbf{m}_r^T}{s^2} \quad (9)$$

where $\mathbf{m}_c$ and $\mathbf{m}_r$ are row vectors containing elements as the sums of the columns and the sums of the rows of the confusion matrix respectively, and s is the sum of the all entries in the confusion matrix.

B. Results and Discussion

Five experiments were performed to evaluate the methods. The overall comparison results of CSP-NBPW, FBCSP-NBPW, Deep ConvNet, C2CM, TSCNN, SF-TSCNN, and

TABLE I
KAPPA AND ACCURACY FOR DIFFERENT METHODS ON THE TEST DATA OF BCI IV 2A DATASET

Method	Kappa (Accuracy)	Subject								
		A01	A02	A03	A04	A05	A06	A07	A08	A09
FBSF-TSCNN	0.627 (0.720)	0.810 (0.858)	0.468 (0.601)	0.838 (0.878)	0.523 (0.642)	0.315 (0.486)	0.426 (0.569)	0.773 (0.830)	0.755 (0.816)	0.736 (0.802)
CSP-NBPW [16]	0.482 (0.611)	0.574 (0.681)	0.366 (0.524)	0.676 (0.757)	0.454 (0.590)	0.130 (0.347)	0.231 (0.424)	0.616 (0.712)	0.593 (0.694)	0.699 (0.774)
FBCSP-NBPW [16]	0.573 (0.680)	0.755 (0.816)	0.449 (0.587)	0.801 (0.851)	0.505 (0.628)	0.273 (0.455)	0.296 (0.472)	0.792 (0.844)	0.671 (0.753)	0.616 (0.712)
Deep ConvNet [28]	0.599 (0.699)	0.792 (0.844)	0.375 (0.531)	0.824 (0.868)	0.500 (0.625)	0.204 (0.403)	0.426 (0.569)	0.824 (0.868)	0.704 (0.778)	0.741 (0.806)
C2CM [27]	0.616 (0.712)	0.769 (0.826)	0.500 (0.625)	0.801 (0.851)	0.537 (0.653)	0.347 (0.510)	0.338 (0.503)	0.773 (0.830)	0.750 (0.812)	0.727 (0.795)
TSCNN ^o	0.172	0.213	0.144	0.213	0.190	0.074	0.167	0.139	0.153	0.255
TSCNN	0.263	0.403	0.146	0.496	0.148	0.088	0.171	0.190	0.259	0.468
SF-TSCNN ^o	0.305	0.412	0.273	0.486	0.199	0.139	0.139	0.375	0.343	0.380
SF-TSCNN	0.552	0.773	0.398	0.755	0.477	0.157	0.310	0.644	0.736	0.722
FBSF-TSCNN ^o	0.350	0.491	0.315	0.514	0.227	0.236	0.176	0.435	0.366	0.394

TABLE II
KAPPA AND ACCURACY FOR DIFFERENT METHODS ON THE TEST DATA OF SMR-BCI DATASET

Method	Kappa (Accuracy)	Subject													
		S01	S02	S03	S04	S05	S06	S07	S08	S09	S10	S11	S12	S13	S14
FBSF-TSCNN	0.657 (0.829)	0.700 (0.850)	0.767 (0.883)	1.000 (1.000)	0.900 (0.950)	0.400 (0.700)	0.367 (0.683)	0.867 (0.933)	0.933 (0.967)	1.000 (1.000)	0.433 (0.717)	0.767 (0.883)	0.633 (0.817)	0.267 (0.633)	0.167 (0.583)
CSP-NBPW [16]	0.495 (0.748)	0.467 (0.733)	0.533 (0.767)	1.000 (1.000)	0.700 (0.850)	0.267 (0.633)	0.300 (0.650)	0.733 (0.867)	0.867 (0.933)	0.933 (0.967)	0.000 (0.500)	0.700 (0.850)	0.400 (0.700)	0.000 (0.500)	0.033 (0.517)
FBCSP-NBPW [16]	0.593 (0.797)	0.433 (0.717)	0.667 (0.833)	1.000 (1.000)	0.833 (0.917)	0.233 (0.617)	0.300 (0.650)	0.833 (0.917)	0.900 (0.950)	0.967 (0.983)	0.433 (0.717)	0.800 (0.900)	0.567 (0.783)	0.200 (0.600)	0.133 (0.567)
Deep ConvNet [28]	0.619 (0.810)	0.533 (0.767)	0.700 (0.850)	1.000 (1.000)	0.900 (0.950)	0.600 (0.800)	0.333 (0.667)	0.933 (0.967)	0.800 (0.900)	0.900 (0.950)	0.333 (0.667)	0.700 (0.850)	0.567 (0.783)	0.233 (0.617)	0.133 (0.567)
C2CM [27]	0.614 (0.807)	0.567 (0.783)	0.767 (0.883)	1.000 (1.000)	0.867 (0.933)	0.433 (0.717)	0.467 (0.733)	0.867 (0.933)	0.833 (0.917)	0.967 (0.983)	0.233 (0.617)	0.767 (0.883)	0.567 (0.783)	0.233 (0.617)	0.033 (0.517)
TSCNN ^o	0.305	0.267	0.333	0.533	0.367	0.167	0.200	0.367	0.267	0.867	0.100	0.267	0.333	0.100	0.100
TSCNN	0.431	0.267	0.533	0.900	0.700	0.267	0.233	0.600	0.267	0.967	0.100	0.567	0.367	0.100	0.167
SF-TSCNN ^o	0.364	0.233	0.367	0.767	0.433	0.133	0.367	0.467	0.500	0.933	0.067	0.467	0.233	0.067	0.067
SF-TSCNN	0.586	0.500	0.667	1.000	0.833	0.333	0.367	0.867	0.933	1.000	0.100	0.767	0.667	0.067	0.100
FBSF-TSCNN ^o	0.474	0.533	0.467	0.767	0.800	0.267	0.367	0.467	0.733	1.000	0.233	0.400	0.367	0.133	0.100

Note: For the methods of TSCNN, SF-TSCNN, and FBSF-TSCNN, the superscript ^o indicates that the TSCNN block of the model adopts the end-to-end training strategy; otherwise, the TSCNN block of the model adopts the stage-wise training strategy.

FBSF-TSCNN on two public datasets with normal settings are shown in Table I and Table II, respectively. Additionally, A graphical representation of the confusion matrix based on Table I is shown in Fig. 3, where the proposed approach FBSF-TSCNN has the larger diagonal elements and smaller non-diagonal elements of the confusion matrix compared with the other methods. Moreover, Table III presents the paired T-tests results among different methods based on Table I and Table II, where all p values were adjusted by false discovery rate (FDR) correction for multiple comparisons.

1) *Results of Experiment I:* To study the impact of different data transformations on the classification performance of the follow-up TSCNN block, we compared the methods of TSCNN, SF-TSCNN, and FBSF-TSCNN.

From Table I, we can see that the SF-TSCNN increases 109.9% higher kappa than the TSCNN due to the employment of spatial filtering, which extracts the dependencies of raw channels and applies the learned spatial filters to channel projection. Fig. 4 shows the topographic map of the most discriminative pair of spatial filters ($N_w = 1$) learned by the SF

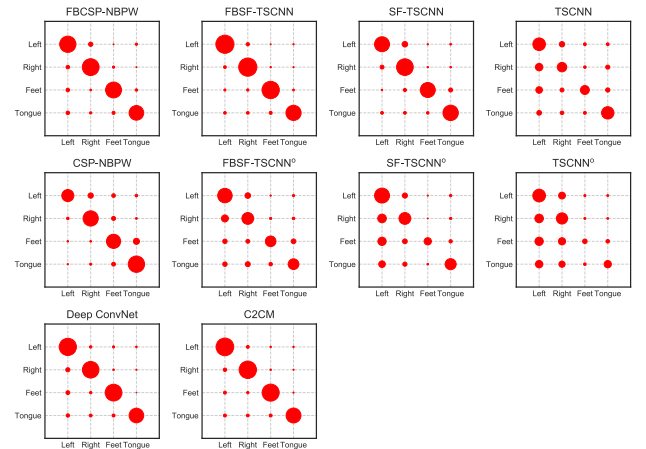


Fig. 3. Confusion matrices of different methods on the BCI IV 2a dataset. The diameter (size) of the circle indicates the magnitude of the corresponding entry in the confusion matrix.

method regarding four-class MI tasks from subject A03 under the OVR strategy. For ease of analysis, the corresponding

TABLE III

PAIRED T-TEST RESULTS BETWEEN KAPPA VALUES OF DIFFERENT METHODS BASED ON TABLE I AND TABLE II

Paired T-test	Adjusted p-value	
	BCI IV 2a dataset	SMR-BCI dataset
FBSF-TSCNN vs. Deep ConvNet	~ (0.146)	~ (0.137)
FBSF-TSCNN vs. C2CM	~ (0.405)	~ (0.067)
FBSF-TSCNN vs. SF-TSCNN	** (5.1e-3)	* (0.032)
SF-TSCNN vs. TSCNN	†† (8.7e-4)	* (0.011)
FBSF-TSCNN vs. FBCSP-NBPW	* (0.016)	* (0.011)
SF-TSCNN vs. CSP-NBPW	* (0.016)	†† (6.0e-4)
FBSF-TSCNN vs. FBSF-TSCNN ^o	†† (3.0e-4)	†† (6.0e-4)
SF-TSCNN vs. SF-TSCNN ^o	†† (8.7e-4)	†† (7.7e-4)
TSCNN vs. TSCNN ^o	* (0.042)	** (8.4e-3)

Note: ~ nonsignificant, * $p \leq 0.05$, ** $p \leq 0.01$, † $p \leq 0.005$, †† $p \leq 0.001$

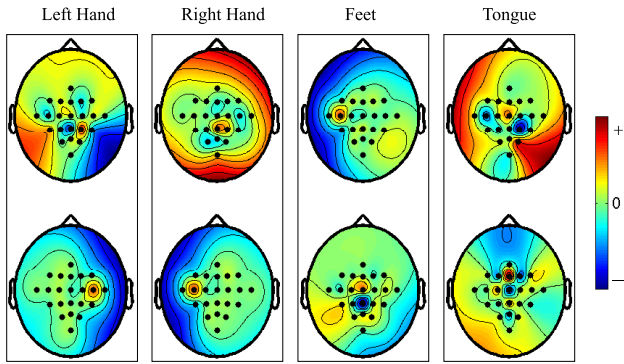


Fig. 4. Topographic maps (of the first and last spatial filters) learned by the SF method regarding four-class MI tasks from subject A03.

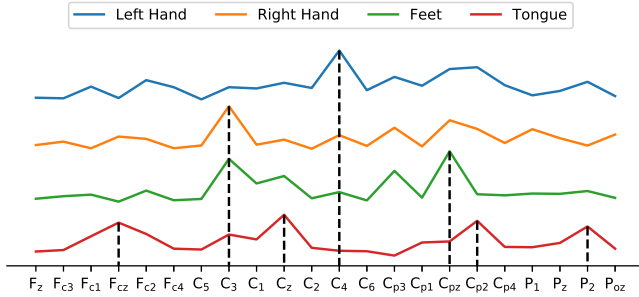


Fig. 5. Visualization of channel weights learned by the SF method regarding four-class MI tasks from subject A03, corresponding to Fig. 4.

average absolute channel weights $\frac{1}{2 \cdot N_w} \sum_{i=1}^{2 \cdot N_w} \left| \frac{\mathbf{w}_i}{\|\mathbf{w}_i\|} \right|$ is shown in Fig. 5, where $\mathbf{w}_i$ is a weight vector of the i^{th} spatial filter, and parameter N_w is set to 1 to be consistent with Fig. 4. We can see that, electrodes C_3 and C_4 have the largest discriminant weight in the imagery movement of the right hand and left hand, respectively. Additionally, electrodes C_3 and C_{p2} have high weights in the feet imagery movements; electrodes C_z , C_{p2} , F_{c2} and P_2 have high weights in the tongue imagery movements. These findings indicate that the discriminant weight of the EEG electrodes varies regarding different MI tasks.

Moreover, the FBSF-TSCNN increases 13.6% higher kappa than the SF-TSCNN because the FBSF block extra performs

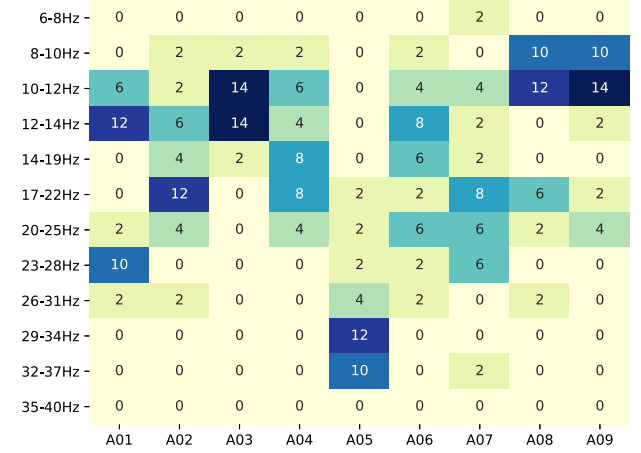


Fig. 6. Results of selecting spatially filtered channels from 12 frequency bands by the FBSF on the BCI IV 2a dataset. The digital in the box represents the number of spatially filtered channels which are selected from the corresponding frequency band for the specific subject.

the selection of spatially filtered channels from multiple frequency bands via MIBIF. Table II demonstrates the same trends on the SMR-BCI dataset. From the paired T-test results shown in Table III, (e.g., FBSF-TSCNN vs. ST-TSCNN, and SF-TSCNN vs. TSCNN), it is clear the FBSF block can improve the classification performance of the follow-up TSCNN block significantly. Although a wide frequency band of 8-30Hz is suggested for manual setting, the research in [16] has proved that selecting subject-specific frequency bands could improve the classification performance of the MI-based BCI. As a typical example, the BCI IV 2a dataset is used to show the details of selecting the spatially filtered channels from 12 frequency bands by the FBSF. For each subject, a total of 32 spatially filtered channels are selected under the OVR strategy, with an average of 8 channels is used to distinguish one class from the rest. As shown in Fig. 6, the results of channel selection vary among different subjects. For all individuals except subject A05, most of the selected (spatially filtered) channels come from the low-frequency bands (i.e., below 31Hz), which indicates that the low-frequency bands are more discriminative for MI decoding.

2) *Results of Experiment II:* To verify whether the time series of spatially filtered EEG contributes to the performance improvement of MI decoding, we compared different methods (e.g., FBSF-TSCNN vs. FBCSP-NBPW, and SF-TSCNN vs. CSP-NBPW) on two public datasets. From Table I, we can see that, the FBSF-TSCNN increases 9.4% higher classification kappa than the FBCSP-NBPW, and the SF-TSCNN increases 14.5% higher classification kappa than the CSP-NBPW. Table II demonstrates the same trends on the SMR-BCI dataset. In Table III, the paired T-test results of FBSF-TSCNN vs. FBCSP-NBPW and SF-TSCNN vs. CSP-NBPW on two public datasets are significant. The reason for this is that the CSP-based methods used the energy (log-variance) representation of the spatially filtered EEG for classification and no longer maintains the representation of time series, thus failing to further exploit the temporal information of spatially filtered EEG. And the numerical results

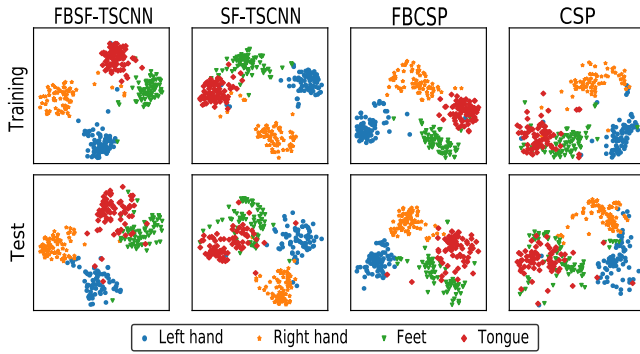


Fig. 7. Comparison of the features from subject A03 learned by different methods in the 2-D embedding space by t-SNE. The top row represents the embedding on the training data, and the bottom row represents the embedding on the test data.

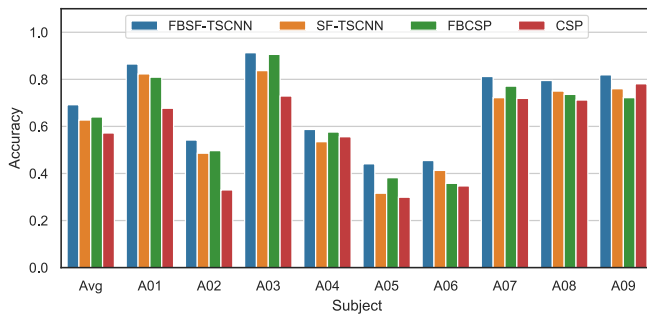


Fig. 8. Testing accuracy comparison of using the NBPW classifier to classify the 2-D embedding of features learned by different methods on the BCI IV 2a dataset.

confirm that the time series of spatially filtered EEG helps to improve MI decoding performance.

Furthermore, to demonstrate the high separability of the features learned by the proposed TSCNN block, we used the t-SNE [40] method to visualize the features learned by different methods in the 2-D embedding space, as shown in Fig. 7. For a fair comparison, we used the NBPW classifier to classify these 2-D embedding on the BCI IV 2a dataset. From the results in Fig. 8, we can see that using the NBPW classifier to classify the 2-D embedding of features learned by the FBSF-TSCNN(SF-TSCNN) yields 8.13% (9.62%) higher accuracy than using the NBPW classifier to classify the 2-D embedding of features learned by the FBCSP(CSP). The results indicate that the features learned by the TSCNN block are more separable than the energy features learned by the CSP-based methods.

It is also important to note that the competing method C2CM [27] used the time series of EEG as the input to the network for MI decoding. The major difference between the FBSF-TSCNN and C2CM are that the FBSF-TSCNN used the spatially filtered EEG as the input to the network, while the C2CM used the envelope of spatially filtered EEG as the input to the network. Although the envelope of EEG signal maintains a temporal representation, it misses some temporal information to some extent compared to the spatially filtered EEG. As shown in Table I and Table II, FBSF-TSCNN

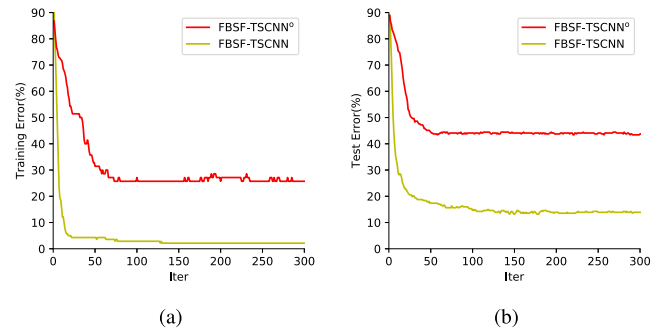


Fig. 9. Comparison of training errors (left) and test errors (right) for subject A03 by the FBSF-TSCNN and FBSF-TSCNN°.

achieves better decoding performance than C2CM, justifying the advantage of the proposed approach.

3) Results of Experiment III: To verify the efficacy of the proposed stage-wise training strategy for the TSCNN block, we compared it with the conventional end-to-end training strategy. From Table I and Table II, it is noticeable that for all subjects, the performance of FBSF-TSCNN achieves higher kappa values than FBSF-TSCNN° on two public datasets. Also, the similar trends are presented in other comparisons, such as SF-TSCNN vs. SF-TSCNN°, and TSCNN vs. TSCNN°. From the paired T-test results in Table III, it is clear that the proposed stage-wise training strategy yields significant performance improvement for the TSCNN block. Fig. 9 shows the training errors and test errors of subject A03 by the methods of FBSF-TSCNN and FBSF-TSCNN°, both with the learning rate of 0.001, betas of (0.9, 0.999), weight penalty of 0.1 and batch size of 100. We can see that the training error of the FBSF-TSCNN is always lower than that of FBSF-TSCNN°, and thus test error is also.

Although some advanced neural modules are drawn into the proposed TSCNN block (for example, adopting the ReLU activation function and batch normalization module to prevent the vanishing gradient problem), the end-to-end trained TSCNN block still has a poor performance. A reasonable explanation is that in the case of insufficient training samples, the TSCNN block is susceptible to poor local minima during gradient-based optimization starting from random initialization [41]. Further, the number of parameters of the feature extraction layers accounts for over 90% of the total number of parameters of the entire TSCNN block. Hence, the advantages of the proposed training strategy are as follows. In the first stage, the weights of the feature extraction layers can be positioned in a region near good local minima by optimization of the triplet loss, which minimizes the intra-class distance and maximizes the inter-class distance collaboratively; in the second stage, only the weights of the classification layers are trained by optimization of the cross-entropy loss, thus alleviating the optimization difficulty of the TSCNN block; in the third stage, the entire network is fine-tuned by the BP algorithm to yield the better generalization.

4) Results of Experiment IV: In this experiment, we aim to compare the effect of different network configurations (*e.g.*, different type of pooling layers, or different dropout percentage

TABLE IV

KAPPA FOR FBSF-TSCNN WITH DIFFERENT DROPOUT PERCENTAGES ON THE TEST DATA OF BCI IV 2A DATASET

Dropout Percentage	Average Kappa	Subject								
		A01	A02	A03	A04	A05	A06	A07	A08	A09
0.3	0.597	0.778	0.412	0.792	0.472	0.343	0.347	0.796	0.727	0.708
0.4	0.612	0.787	0.435	0.829	0.509	0.324	0.384	0.778	0.736	0.727
0.5	0.627	0.810	0.468	0.838	0.523	0.315	0.426	0.773	0.755	0.736
0.6	0.609	0.796	0.449	0.824	0.519	0.306	0.389	0.727	0.741	0.732
0.7	0.569	0.671	0.435	0.792	0.486	0.282	0.357	0.634	0.741	0.727
0.8	0.470	0.644	0.352	0.713	0.218	0.278	0.107	0.611	0.685	0.625

TABLE V

KAPPA FOR DIFFERENT METHODS ON THE TEST DATA OF BCI IV 2A DATASET WITH 1/2 TRAINING DATA

Method	Average Kappa	Subject								
		A01	A02	A03	A04	A05	A06	A07	A08	A09
FBSF-TSCNN	0.578	0.750	0.403	0.824	0.510	0.347	0.241	0.769	0.699	0.662
C2CM [27]	0.569	0.713	0.417	0.815	0.519	0.319	0.296	0.681	0.694	0.667
Deep ConvNet [28]	0.556	0.750	0.227	0.819	0.486	0.245	0.231	0.801	0.718	0.727
FBCSP-NBPW [16]	0.536	0.713	0.403	0.806	0.509	0.287	0.213	0.694	0.620	0.579
SF-TSCNN	0.523	0.718	0.394	0.718	0.431	0.176	0.255	0.722	0.639	0.657
CSP-NBPW [16]	0.459	0.574	0.384	0.630	0.389	0.134	0.227	0.653	0.560	0.574

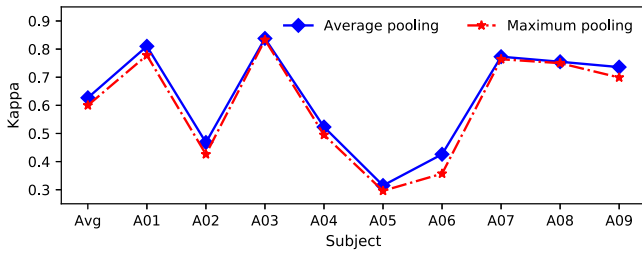


Fig. 10. Comparison of classification kappa for FBSF-TSCNN with different type of pooling layers in the TSCNN block.

settings) on the decoding results of the proposed TSCNN block.

As shown in Fig. 10, for the proposed approach FBSF-TSCNN, if the average pooling layer of the TSCNN block is replaced by the maximum pooling layer, then the average kappa value decreases from 0.627 to 0.600. It is important to note that the feature maps input to the pooling layer are extracted sequentially from the previous three convolutional layers, so they have high-level abstraction to some extent. However, the maximum pooling layer selects only the maximum element within the receptive field as the output, ignoring the contributions of other elements within the receptive field. In contrast, the average pooling layer takes into account all elements within the receptive field. This also explains why the average pooling layer yields the better decoding results than the maximum pooling layer in the proposed approach.

In general, the dropout percentage (probability of a unit to be zeroed) is set to a default value of 0.5, which seems to be close to optimal for a wide range of networks and tasks [42]. In order to select the optimal dropout percentage for the proposed approach, we trained the network with different dropout percentages (e.g., 0.3, 0.4, 0.5, 0.6, 0.7, 0.8) on BCI IV 2a dataset. As shown in Table IV, we can see that, the network trained with dropout percentage of 0.5 performs best, achieving the highest average kappa of 0.627 on the test

data of BCI IV 2a dataset. Moreover, if the dropout percentage is closer to 0.5, the trained network has better classification performance; otherwise, if the dropout percentage is far from 0.5, the classification performance of the network is consequently worse.

5) *Results of Experiment V:* To test the performance of the proposed approach for a small training dataset, we only used 1/2 (i.e., a total of 144 trials with an average of 36 trials for each class) training data of BCI IV 2a dataset, and compared it with other methods. The classification results are reported in Table V. Compared with the results in Table I, the classification performance of all methods decreased (degradation ratio: FBSF-TSCNN (7.8%), C2CM(8.3%), Deep ConvNet (7.2%), FBCSP-NBPW (6.9%), SF-TSCNN (5.3%), and CSP-NBPW (4.8%)). Although the proposed approach has slightly higher performance degradation, it still achieves the highest average kappa value of 0.578. The p-values of paired T-tests of FBSF-TSCNN with other methods (except Deep ConvNet and C2CM) are smaller than 0.05, which reveals that the proposed approach could work well in a small dataset case (1/2 size setting).

IV. ANALYSIS

A. Complexity of the Deep Models

The complexity of the proposed approach was compared with the state-of-the-art methods Deep ConvNet and C2CM. On the BCI IV 2a dataset (SMR-BCI dataset), the Deep ConvNet has up to 2.79×10^5 (2.73×10^5) parameters, the C2CM has 4.9×10^3 (3.9×10^3) parameters, and the FBSF-TSCNN has 1.03×10^4 (9.01×10^3) parameters. The parameter complexity of FBSF-TSCNN is much less than Deep ConvNet, but slightly greater than C2CM. In general, the compact model with fewer parameters to be optimized has better robust generalizability to the test set than the complicated one [28]. Therefore, the proposed approach, which has the highest classification performance and small complexity, is promising in the applications of the MI-based BCI system.

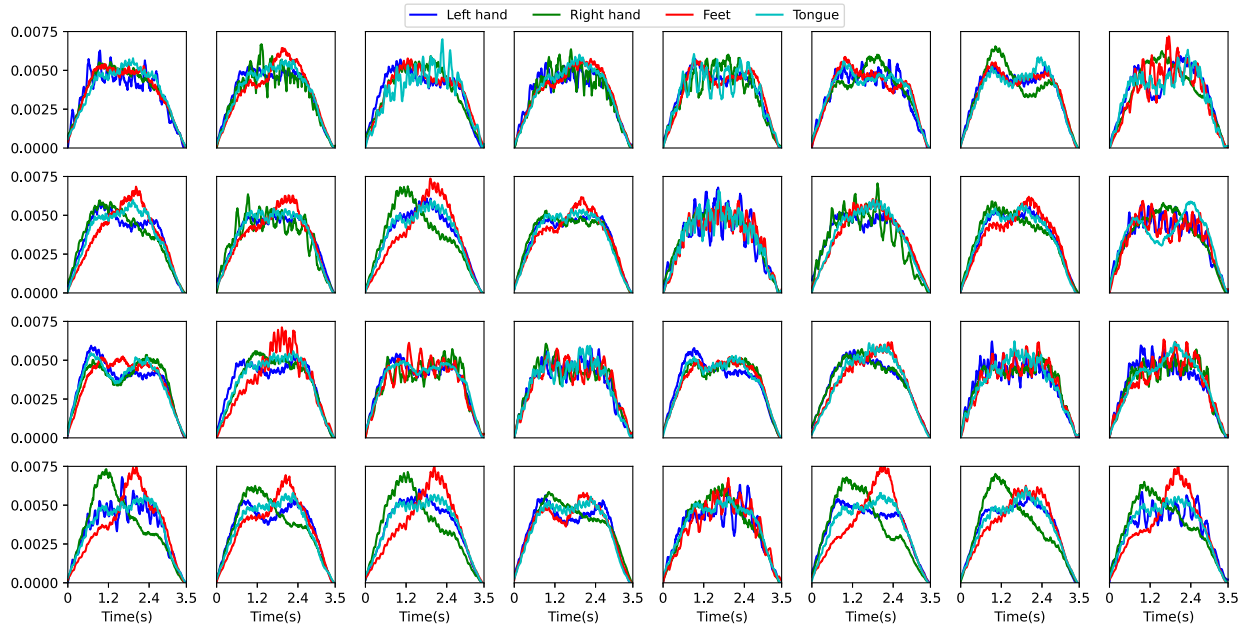


Fig. 11. The class-specific sensitivity analysis result of 32 spatially filtered EEG channels for subject A03, where the inputs of the TSCNN block are the intermediate EEG signals provided by the FBSF block.

B. Interpretation of Discriminative Temporal Information

Apart from improving the kappa value of motor imagery decoding, it is also important to interpret the discriminative temporal information learned by the proposed approach from EEG signals. As a typical example, subject A03 is selected to analyze the sensitivity of the proposed TSCNN block to input signals. For the intermediate EEG trial X_i provided by the FBSF block, the element at time t from the time-series of one channel is denoted as e_t . According to the research in [43], the sensitivity of the model M to the element e_t is defined as the partial derivatives of M to e_t as follows:

$$S(e_t) = \left| \frac{\partial M(X_i; \theta)}{\partial e_t} \right| \quad (10)$$

Then, for each class of MI task, the average sensitivity of the element e_t of different channels is calculated. Fig. 11 shows the analysis results of the proposed TSCNN block for subject A03. It can be seen that for each class of MI task, the time series of the intermediate period from all spatially filtered channels are more sensitive to the final decoding results, which reveals that the ERD/ERS pattern in the intermediate period is more discriminative than other periods, and thus the proposed approach mainly exploits the temporal information of the intermediate period. A reasonable explanation for the analysis results is that the subject tends to concentrate on performing MI tasks in the intermediate stage, while they may not be ready or feel distracted in the beginning and ending stages. Therefore, the temporal information in the intermediate stage is more conducive to MI decoding [44].

C. Limitation and Future Directions

Though the proposed FBSF-TSCNN has achieved cutting-edge performance in motor imagery decoding, the limitation

is still involved in the current work. For the TSCNN block, the feature extraction layers and the classification layers are trained separately under the stage-wise training strategy. Therefore, in the future work, we will focus on the end-to-end training of the whole TSCNN block by constructing a new optimization objective that combines the triplet loss and cross-entropy loss, and also further explore the potential of the proposed approach.

V. CONCLUSION

In summary, temporal information of EEG signals is essential to the performance improvement of motor imagery decoding. However, the existing EEG feature extraction methods, such as CSP or FBCSP, ignore the further utilization of the temporal information. In this paper, we propose an approach FBSF-TSCNN that combined filter bank spatial filtering and deep temporal-spatial feature learning. Firstly, the FBSF block transforms the raw EEG signals into an appropriate intermediate EEG representation, which preserves the discriminative information and eliminates components that are not related to the classification. Then, the TSCNN block is proposed to learn the discriminative features of the intermediate EEG signals and return the decoding results. A novel stage-wise training strategy is developed to optimize the TSCNN block, which yields significant performance improvement compared with the conventional end-to-end training strategy. The experimental results on the BCI IV 2a and SMR-BCI datasets show the cutting-edge performance of the proposed approach. Therefore, the proposed approach is a promising tool for the research and application in MI-based BCI systems.

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